

Analysis Matrix quiz 1 2025-26

Please State Your I _____

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 6 & -2 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 3 \\ -1 & 1 \\ 1 & -1 \end{bmatrix}$$

For questions 1-6, reference the matrices above. Perform the given operations or state that it does not exist (DNE) [2 points each]

1. $4B - C$

2. D^T

3. BA

4. CD

5. DA

6. Using the matrix C above, create 2 different matrices E (besides $E = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$) such that $CE=EC$.

[2]

7. Solve the system of equations using matrices. Clearly show all your steps. [4]

$$2x + y = 19$$

$$x - y = -1$$

Final Answer: _____

8. Given the matrices $A = \begin{bmatrix} 2 & 4 \\ 2 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ -1 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 6 & 2 \\ 4 & 1 \end{bmatrix}$, solve for the matrix X in the equation $X(A - 5B) = C$. [4]

Simplify your answer.

9. Solve for both values of x that would make the determinant of matrix $F = 0$. [4]

$$F = \begin{bmatrix} x & 1 & 0 \\ 2 & x & 1 \\ 0 & 3 & 4 \end{bmatrix}$$